Exam 2 (solved)

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Question 1

Suppose q_1, q_2, q_3 are orthonormal vectors in $\mathbb{R}^3$. Find all possible values for these 3×3 determinants and explain your thinking in 1 sentence each.

(a) $\det[q_1\ q_2\ q_3]$

Since q_1, q_2, q_3 are orthonormal, the matrix $Q = [q_1\ q_2\ q_3]$ is orthogonal, so its determinant must be $+1$ or -1.

Step 1: q_1, q_2, q_3 are orthonormal, so the matrix $Q = [q_1\ q_2\ q_3]$ is an orthogonal matrix.

Step 2: The determinant of any orthogonal matrix in $\mathbb{R}^3$ is either $+1$ or -1 (see Determinants.ipynb).

Conclusion: $\det[q_1\ q_2\ q_3] = \pm 1$.

(b) $\det[q_1 + q_2,\ q_2 + q_3,\ q_3 + q_1]$

By multilinearity, expanding the determinant gives a sum where only the terms with all columns different survive, which are cyclic permutations of (q_1, q_2, q_3). Each such term is ± 1, and there are two of them, so the answer is ± 2.

Step 1: Let $A = [q_1 + q_2,\ q_2 + q_3,\ q_3 + q_1]$.

Step 2: By the linearity of the determinant in each column, expand:

$$\det[q_1 + q_2,\ q_2 + q_3,\ q_3 + q_1] = \det[q_1, q_2, q_3] + \det[q_1, q_3, q_1] + \det[q_2, q_2, q_3] + \dots$$

But most terms vanish because two columns are equal (determinant is zero if two columns are equal).

Step 3: The only nonzero terms are those corresponding to permutations of (q_1, q_2, q_3).

$$\det A = \det[q_1, q_2, q_3] + \det[q_2, q_3, q_1]$$

(This is derived by applying $C_1' = C_1 + C_2$, $C_2' = C_2 + C_3$, $C_3' = C_3 + C_1$, and using linearity and elimination of zero terms).

Step 4: There are three such terms when fully expanded using $2^3 = 8$ terms. The result is $2\det[q_1, q_2, q_3]$.

Step 5: Since $\det[q_1, q_2, q_3] = \pm 1$, the resulting determinant is $2(\pm 1)$.

Conclusion: $\det[q_1 + q_2, q_2 + q_3, q_3 + q_1] = \pm 2$.

(c) $\det[q_1\ q_2\ q_3] \times \det[q_2\ q_3\ q_1]$

Cyclically permuting the columns of an orthogonal matrix preserves the sign of the determinant, so both determinants are equal (± 1), and their product is always 1.

Step 1: $\det[q_1\ q_2\ q_3]$ is ± 1.

Step 2: $\det[q_2\ q_3\ q_1]$ involves a cyclic permutation of the columns $(1 \to 2 \to 3 \to 1)$, which corresponds to two swaps (an even permutation), so the sign stays the same: $\det[q_2\ q_3\ q_1] = \det[q_1\ q_2\ q_3]$.

Step 3: So the product is $(\pm 1) \times (\pm 1) = 1$.

Conclusion: $\det[q_1\ q_2\ q_3] \times \det[q_2\ q_3\ q_1] = 1$.

Question 2

Suppose we take measurements at the 21 equally spaced times $t = -10, -9, \ldots, 9, 10$. All measurements are $b_i = 0$ except that $b_{11} = 1$ at the middle time $t = 0$.

(a) Using least squares, what are the best C and D to fit those 21 points by a straight line $C + Dt$?

The model is $b_i \approx C + Dt_i$. Since the times are symmetric about zero and only the middle value is nonzero, the best fit line is horizontal: $D = 0$. The sum of the fitted values must equal the sum of the data, so $C = 1/21$.

Step 1: Write the model as $b_i \approx C + Dt_i$. The normal equations are $A^T A x = A^T b$.

Step 2: Since the times are symmetric around zero, the cross-term in $A^T A$ is $\sum t_i = 0$.

$$A^T A = [21 \quad 0\, 0 \quad 770]$$

Step 3: Compute $A^T b$. Since b_i is nonzero only at $t = 0$:

$$A^T b = [\sum 1 \cdot b_i\ \sum t_i \cdot b_i] = [1\ 0]$$

Step 4: Solve for $x = [C\ D]$:

$$[21 \quad 0\, 0 \quad 770]\, [C\ D] = [1\ 0]$$

$C = \frac{1}{21}$, $D = 0$.

Conclusion: The best fit is $C = \frac{1}{21}$, $D = 0$.

(b) You are projecting the vector b onto what subspace? (Give a basis.) Find a nonzero vector perpendicular to that subspace.

The subspace is the span of the constant vector $[1, 1, \ldots, 1]^T$ and the time vector $[t_1, \ldots, t_{21}]^T$. A vector perpendicular to both is $[1, -2, 1, 0, \ldots, 0]^T$ (a discrete second difference), since it sums to zero and is orthogonal to the time vector. The error vector $e = b - P_A b$ is also perpendicular to the subspace.

Step 1: The subspace is the column space of A, spanned by the constant vector $\mathbf{1}$ and the time vector $\mathbf{t}$.

Basis: $\mathbf{1}, \mathbf{t}$, where $\mathbf{1} = [1, 1, \ldots, 1]^T$ and $\mathbf{t} = [-10, -9, \ldots, 10]^T$.

Step 2: A vector perpendicular to this subspace must be orthogonal to both $\mathbf{1}$ and $\mathbf{t}$. The vector b itself, where $b = [0, \ldots, 0, 1, 0, \ldots, 0]$, is highly suggestive.

Step 3: However, we want a general vector perpendicular to *any* line. A vector v that has zero sum (orthogonal to $\mathbf{1}$) and is symmetric (orthogonal to $\mathbf{t}$) works. A simpler standard orthogonal vector is related to the discrete second derivative (measures curvature, which a straight line does not possess).

Step 4: Consider v such that $v_i = 1$ for $i = 1$, $v_i = -2$ for $i = 2$, $v_i = 1$ for $i = 3$, and $v_i = 0$ otherwise. This v is 21×1. This v has $\sum v_i = 1 - 2 + 1 = 0$. If the times t_i start at $i = 1$, then $\sum v_i t_i = 1 \cdot t_1 - 2 \cdot t_2 + 1 \cdot t_3$. Since $t_2 = t_1 + 1$ and $t_3 = t_1 + 2$, this sum is $t_1 - 2t_1 - 2 + t_1 + 2 = 0$.

Conclusion: The subspace is spanned by $[1, 1, \ldots, 1]^T$ and $[t_1, \ldots, t_{21}]^T$. A nonzero vector perpendicular to it is $v = [1, -2, 1, 0, \ldots, 0]^T$ (assuming the first three measurements correspond to $t = -10, -9, -8$).

Question 3

The Gram-Schmidt method produces orthonormal vectors q_1, q_2, q_3 from independent vectors a_1, a_2, a_3 in $\mathbb{R}^5$. Put those vectors into the columns of 5×3 matrices Q and A.

(a) Give formulas using Q and A for the projection matrices P_Q and P_A onto the column spaces of Q and A.

The projection matrix onto the column space of Q is $P_Q = QQ^T$, and for A it is $P_A = A(A^T A)^{-1} A^T$.

Conclusion:

- $P_Q = QQ^T$ (Since $Q^TQ = I$)
- $P_A = A(A^TA)^{-1}A^T$

(b) Is $P_Q = P_A$ and why? What is P_Q times Q? What is $\det P_Q$?

Since Q is obtained from A by Gram-Schmidt, both have the same column space, so $P_Q = P_A$. Multiplying P_Q by Q gives Q back. P_Q is a 5×5 projection matrix of rank 3, so it is singular and $\det P_Q = 0$.

Conclusion:

- $P_Q = P_A$ because both project onto the same subspace (Gram-Schmidt preserves the column space).
- $P_Q Q = QQ^TQ = Q(Q^TQ) = QI = Q$.
- $\det P_Q = 0$ because P_Q is a 5×5 matrix with rank 3, meaning its null space is non-trivial, so it is singular.

(c) Suppose a_4 is a new vector and a_1, a_2, a_3, a_4 are independent. Which of these (if any) is the new Gram-Schmidt vector q_4? (Use P_A and P_Q from above)

Step 1: Gram-Schmidt requires subtracting the projection of the new vector (a_4) onto the previously spanned space ($\mathrm{Col}(A)$ or $\mathrm{Col}(Q)$).

Step 2: Both $a_4 - P_A a_4$ and $a_4 - P_Q a_4$ represent the component of a_4 orthogonal to the subspace spanned by a_1, a_2, a_3.

Step 3: The Gram-Schmidt vector q_4 is the normalized version of this result. Since the question asks "Which of these is the new Gram-Schmidt vector q_4?", it implies the orthogonal vector before normalization.

Conclusion: Therefore choice 3 is the correct definitions for the new orthogonal vector (which must then be normalized to find q_4).

Question 4

Suppose a 4×4 matrix has the same entry $\times$ throughout its first row and column. The other 9 numbers could be anything.

$$A = [\times \quad \times \quad \times \quad \times \times \quad * \quad * \quad * \times \quad * \quad * \quad * \times \quad * \quad * \quad *]$$

(a) The determinant of A is a polynomial in $\times$. What is the largest possible degree of that polynomial? Explain your answer.

In the determinant expansion, you can pick at most one $\times$ from the first row and one from the first column in any term, so the highest possible degree is 2.

Step 1: The determinant is a sum of $n!$ products, each picking one entry per row and column.

Step 2: Any term in the expansion can pick at most one entry from the first row and at most one entry from the first column.

Step 3: The highest degree occurs when we maximize the number of $\times$'s chosen. Case 1: We choose $a_{11} = \times$. This uses one $\times$. The remaining 3×3 submatrix has no $\times$'s, so the term contributes $\times^1$. Case 2: We choose $a_{1j} = \times$ ($j > 1$) and $a_{i1} = \times$ ($i > 1$). This is impossible for the same term, as they share either a row or a column.

Step 4: The maximum degree is achieved by picking the entries a_{11} and two more entries from the first row/column that do not conflict. Wait, that's impossible.

Step 5: Let's re-examine the definition of the determinant: $\sum_\sigma (\text{sgn}(\sigma)) a_{1,\sigma(1)} a_{2,\sigma(2)} a_{3,\sigma(3)} a_{4,\sigma(4)}$. The maximum number of factors of $\times$ per product is 2:

1. Choose $a_{1,1} = \times$. Remaining three factors must be from the 3×3 block of $*$s. Contribution: $\times^1$.
2. Choose $a_{1,j} = \times$ ($j \neq 1$) AND choose $a_{i,1} = \times$ ($i \neq 1$). The remaining 2×2 block of $ *syieldstwofactors. Contribution* :|times^2. (*Example* :a_{12} a_{21} a_{33} a_{44}*ifa*_{33}, a_{44}*arethe*$s).

Conclusion: The largest possible degree of the determinant as a polynomial in $\times$ is 2. (The claim of 3 in the original broken solution was incorrect based on cofactor expansion rules.)

(b) If those 9 numbers give the identity matrix I, what is $\det A$? Which values of $\times$ give $\det A = 0$?

If the $*$ entries form the 3×3 identity, then A is

$$[x \quad x \quad x \quad x\, x \quad 1 \quad 0 \quad 0\, x \quad 0 \quad 1 \quad 0\, x \quad 0 \quad 0 \quad 1]$$

The determinant is $x(1 - 3x)$, so $\det A = 0$ when $x = 0$ or $x = 1/3$.

Step 1: Write out the matrix:

$$A = [x \quad x \quad x \quad x\, x \quad 1 \quad 0 \quad 0\, x \quad 0 \quad 1 \quad 0\, x \quad 0 \quad 0 \quad 1]$$

Step 2: Calculate $\det A$. We can use the formula for a partitioned matrix $[A_{11} \quad A_{12}\, A_{21} \quad A_{22}]$. Here $A_{11} = x$ (scalar), $A_{22} = I$ (3×3), $A_{12} = [x, x, x]$, and $A_{21} = [x, x, x]^T$.

$$\det A = \det(A_{22}) \cdot \det(A_{11} - A_{12} A_{22}^{-1} A_{21})$$

Since $A_{22} = I$, $A_{22}^{-1} = I$.

$$A_{12} I A_{21} = [x \quad x \quad x]\,[x\ x\ x] = x^2 + x^2 + x^2 = 3x^2$$

Thus, $\det A = 1 \cdot (x - 3x^2) = x(1 - 3x)$.

Step 3: Find values of x such that $\det A = 0$. Setting $x(1 - 3x) = 0$, we find $x = 0$ or $1 - 3x = 0$, which gives $x = 1/3$.

Conclusion: $\det A = x(1 - 3x)$, and $\det A = 0$ when $x = 0$ or $x = 1/3$.